

SPECIAL FEATURE: ORIGINAL ARTICLE

Yoshinari Tanaka

Extinction of populations by inbreeding depression under stochastic environments

Received: May 10, 1999 / Accepted: November 5, 1999

Abstract Inbreeding depression may induce rapid extinction due to positive feedbacks between inbreeding depression and reduction of population size, which is often referred to as extinction vortex by inbreeding depression. The present analysis has demonstrated that the extinction vortex is likely to happen with realistic parameter values of genomic mutation rate of lethals or semilethals, equilibrium population size, intrinsic rate of natural increase, and rate of population decline caused by nongenetic extrinsic factors. Simulation models incorporating stochastic fluctuations of population size further indicated that extinction by inbreeding depression is facilitated by environmental fluctuations in population size. The results suggest that there is a positive interaction between genetic stochasticity and environmental stochasticity for extinction of populations by inbreeding depression.

Key words Extinction vortex · Deleterious mutation · Environmental stochasticity · Genetic load · Conservation biology

Introduction

Loss of genetic heterozygosity is considered to be one of the most important factors of extinction for small populations (Frankel and Soule 1981; Caughley and Gunn 1996). Inbreeding depression in particular, which results from an increase in homozygotes of recessive deleterious mutations, may induce rapid extinction of populations as the result of positive interaction between decline in population size and increased inbreeding (the extinction vortex phenomenon, which is simply referred to as inbreeding vortex in this

paper) (Gilpin and Soulé 1986; Tanaka 1997, 1998). Rapid extinction by inbreeding depression requires an equilibrium population to be perturbed by nongenetic external causes and that the population size before the perturbation is large enough to maintain recessive deleterious mutations in the population (Tanaka 1997, 1998). The fitness–heterozygosity correlation and the rate of inbreeding depression estimated for natural populations provide indirect evidences of inbreeding depression in nature (Mitton and Grant 1984; Zouros 1987; Barrett and Charlesworth 1991; Mitton 1993; Willis 1993; Ouborg and Treuren 1994; Britten 1996; David et al. 1997; Schierup 1998). Although data have accumulated on genetic heterozygosity of endangered species (Loeschke et al. 1994; Avise and Hamrick 1996), biological interpretation of the estimates or its importance for the extinction process of populations is not clear.

Lack of information about two aspects causes difficulties. First, we have very few established examples of population extinction caused by inbreeding depression (but see Saccheri et al. 1998). A field survey on Glanville fritillary butterflies (*Melitaea cinxia*) is the only study that has reported a contribution of inbreeding depression to extinction of natural populations (Saccheri et al. 1998). Second, lack of investigations on the theoretical properties of the extinction vortex by inbreeding depression makes it impossible to evaluate genetic risk of extinction from empirical data. My previous studies examined the possibility of an extinction vortex with population genetic and population ecological models, and concluded that rapid extinction caused by inbreeding depression is restricted but realistic with observed genetic parameters (Tanaka 1997, 1998).

Semiquantitative conditions required for the extinction vortex to happen are as follows. The equilibrium population size must be large ($>10^5$) so that recessive deleterious genes are maintained at relatively high frequency. Inbreeding depression cannot alone induce extinction (the genetic and demographic equilibria are locally stable). Nongenetic demographic disturbances are required to reduce the population size at any rates higher than that of purging selection, which eliminates the deleterious genes and makes inbreeding depression irrelevant. Also, the maximum (mutation-

Y. Tanaka
Laboratory of Theoretical Ecology, Institute of Environmental
Science and Technology, Yokohama National University, 79-7
Tokiwadai, Hodogaya, Yokohama 240-8501, Japan
Tel. +81-45-339-4367; Fax +81-45-339-4367
e-mail: tanaka@kan.ynu.ac.jp

free) intrinsic rate of natural increase is not high, so that reduction of population size provides positive feedback through the action of inbreeding depression (Tanaka 1997, 1998). The overdominant genes are very unlikely to contribute to the extinction vortex by inbreeding depression because very high equilibrium segregation loads must precede the inbreeding vortex (Tanaka 1998).

Real populations are subject to random fluctuations of environmental factors such as temperature, food level, and predation pressure. These factors bring about random fluctuations of population size (environmental stochasticity). Extinction vortex caused by inbreeding depression, which is likely to occur in declining populations suffering demographic disturbances (Tanaka 1997, 1998), may be furthermore reinforced or enhanced by the environmental stochasticity of population size (van Noordwijk 1994). The present study examines the possibility of interaction between inbreeding depression and the environmental stochasticity of population size in the process of extinction by inbreeding depression.

Model

I assumed recessive deleterious genes for the genetic mechanism of inbreeding depression. Also, the deleterious genes are assumed to be distributed among n diallelic autosomal loci, which are identical concerning per locus mutation rate, selection coefficient, and degree of dominance. Then, the model is a simple extension of a one-locus, two-alleles model. Linkage disequilibrium and epistatic interaction between loci are disregarded for simplicity. It is assumed that three genotypes (AA, Aa, and aa) have mean fitnesses as 1, 1, and $1 - s$, respectively. Thus, all deleterious genes are assumed to be completely recessive. Incomplete dominance of the wild-type allele must reduce inbreeding depression and make the inbreeding vortex less likely to happen.

Mean fitness of a population decreases due to deleterious genes from the maximum value 1 by L . This value is called genetic load in population genetics (Crow and Kimura 1970; Nei 1987). To evaluate the effects of inbreeding depression on extinction of populations, the genetic load must be translated into demographic parameters. I assumed that Malthusian fitness ($W = e^r$) decreases linearly with the genetic load. Because the realized population growth rate is $e^{r(1-N/K)}$ under density-dependent effects, the population growth rate or the mean fitness of a genetically loaded population is $\lambda = (1 - L)e^{r_{\max}(1-N/K)}$ where $r_{\max}$ is the maximum intrinsic rate of population increase of a mutation-free population.

A population is assumed to be at demographic and genetic equilibria. Demographic disturbances caused by anthropogenic factors (e.g., destruction of habitats and hunting) cause reduction of population size from the equilibrium value. A previous study has suggested that the equilibria are locally stable with realistic parameter values (Tanaka 1998). If the reductions of population size diminish

over time, the population reaches a new equilibrium size and will not go extinct. However, if the reductions are amplified over time, the population will quickly go extinct by inbreeding vortex (see later sections).

At equilibrium, the expected inbreeding coefficient and the mean gene frequency over loci are kept unchanged by a balance between mutation, genetic drift, and selection. The genetic load is also at equilibrium. The population growth rate at equilibria is denoted as $\tilde{\lambda} = (1 - \tilde{L})e^{r_{\max}(1-N/K)}$, where the tildes represent equilibrium values. By inbreeding depression, the population growth rate reduces by a certain amount. Denote the proportional reduction of λ as $1 - \delta_{\text{inb}}$; the population growth rate of any inbred population is $\lambda = \tilde{\lambda}\delta_{\text{inb}}$, where $\delta_{\text{inb}} = \frac{1 - L}{1 - \tilde{L}}$.

The effect of the equilibrium genetic load on the population growth rate is disregarded in this analysis so that the contribution of inbreeding depression that is newly generated from demographic disturbances to the extinction process is exclusively incorporated into the analysis. The equilibrium genetic load (and the reproductive rate at equilibria) takes an assumed quantity.

The per locus genetic load at the i th locus l_i is defined as $l_i = 1 - \bar{w}_i/w_{\max}$, where $\bar{w}_i$ is the marginal mean fitness of the i th locus and $w_{\max}$ is the maximum fitness of a locus, which is assumed to be unity for all loci. Assuming the multiplicative fitness without epistatic interaction between loci, the total genetic load L is calculated as

$$\begin{aligned} L &= 1 - \prod_i (1 - l_i) \\ &= 1 - \prod_i (\bar{w}_i)/w_{\max} \end{aligned} \quad (1)$$

where $w_{\max}$ is the maximum multilocus fitness, $\prod_i w_{\max}$, and equivalent to unity. Then, $L = 1 - \prod_i \bar{w}_i$. The marginal mean fitness at the i th locus is $\bar{w}_i = 1 - sq_i^2$, where q_i is the gene frequency of the recessive deleterious gene at the i th locus. Then, the total genetic load is approximately $L \cong s \sum_i q_i^2 = ns E[q_i^2]$, where E denotes the expectation over all loci.

Changes in gene and genotypic frequencies

Gene frequencies of the deleterious genes change by mutation, selection, and genetic drift. Through the action of random genetic drift, gene frequencies disperse between loci. For simplicity, the joint dynamics of the gene frequencies are summarized as changes in the mean gene frequency over all contributing loci, $\bar{q} = n^{-1} \sum_i q_i$, and the variance of the gene frequencies. From $L \cong ns E[q_i^2]$, the total genetic load, which expresses the net effect of inbreeding depression, is largely determined by the first two moments of the distribution of gene frequencies.

The per generation change of gene frequency at a locus by selection is derived from Wright's formula,

$$\Delta_s q_i = \frac{q_i(1 - q_i)}{2} \frac{\partial \ln \bar{w}_i}{\partial q_i}, \text{ in which } \frac{\partial \ln \bar{w}_i}{\partial q_i} \cong -2sq_i. \text{ If all}$$

loci are approximately at linkage equilibria, the change in the mean gene frequency by selection is calculated as

$$\Delta_s \bar{q} = E[\Delta_s q_i] = E[-sq_i^2(1 - q_i)] \cong -sE[q_i^2] \quad (2)$$

The per generation change of the mean gene frequency by mutation is equivalent to the per locus, per gamete mutation rate, $\Delta_m \bar{q} = \mu$, if the mutation is irreversible. Most deleterious mutations that have large adverse effects on fitness are likely to be irreversible.

By random mating in a finite population, gene frequencies starting from an identical initial frequency tend to disperse between independent sets of samplings or stochastic processes (Crow and Kimura 1970; Falconer 1989). The dispersion of gene frequencies is readily expressed by the variance of gene frequencies monotonically increasing with generations, V_q , and the inbreeding coefficient F . If the dispersion is independent, the variance of gene frequencies is expressed as $V_q = F\bar{q}(1 - \bar{q})$ (Crow and Kimura 1970; Falconer 1989). The theory of random dispersion of gene frequencies has been successfully applied to genetic differentiation at a locus between local populations (Crow and Kimura 1970; Nei 1987). If there is no gametic correlation between loci, the random dispersion of gene frequencies holds for different loci within a genome. I employed this approximate treatment for describing changes in gene and genotypic frequencies by genetic drift. The standard theory of inbreeding indicates

$$E[q_i^2] = \bar{q}^2 + V_q = \bar{q}^2 + F\bar{q}(1 - \bar{q}) \quad (3)$$

The inbreeding coefficient changes mostly by inbreeding and partly by selection and mutation. The per generation change in F is expressed by the following recurrence equation: $F_{t+1} = [1/(2N) + (1 - 1/(2N))F_t] (1 - 2\mu)(1 - s\bar{q})$ (Tanaka 1997, 1998).

Equilibrium population

Without demographic disturbance and environmental stochasticity, the population is kept at a demographic and genetic equilibrium in which the population size, the mean gene frequency, and the inbreeding coefficient do not change. I employed these equilibrium values for the initial values of F and $\bar{q}$ in the simulations. It is envisaged that the population had been at a long-term equilibrium before anthropogenic factors started continually degrading populations and disrupting the equilibria.

The long-term effective population size of a stochastically fluctuating population is smaller than the census population size. If there is no autocorrelation in the fluctuations, the effective population size N_e is $N - \sigma_N^2/N$, where σ_N^2 is the variance of population size (Crow and Kimura 1970). If the variance of population size is mostly explained by environmental fluctuation of the population growth rate, the variance of population size is equivalent to $\sigma_N^2 = (\sigma_e^2 K^2 + K)/$

(2r), where σ_e^2 is the environmental variance of the population growth rate, $r\{1 - (N/K)\}$ (Iwasa 1998). For numerical evaluation of equilibrium values of inbreeding coefficient and mean gene frequency, the effective population size was used in place of the population size.

The equilibrium mean gene frequency that is achieved by the balance between mutation and selection ($\Delta_s \bar{q} + \Delta_m \bar{q} = 0$) must meet the following quadratic equation, $(\bar{F} - 1)\bar{q}^2 - \bar{F}\bar{q} + \mu/s \cong 0$, where the tildes represent equilibrium values. It is not possible to find a simple analytical solution of the equilibrium mean gene frequency. The numerical simulations in this study employed $\tilde{q}$ values numerically evaluated with $\Delta_s \bar{q} + \Delta_m \bar{q} = 0$.

The standard protocol of local stability analysis indicates that the demographic and genetic equilibria are locally stable regardless of equilibrium population size, mutation rate, selection coefficient, and number of loci (Tanaka 1998).

Demographic disturbances

Extinction by inbreeding depression occurs only when the population reduces below the demographic and genetic equilibria. Such reductions of population size result from continual demographic disturbances. Destruction of habitats or overhunting can be real factors of demographic disturbances (Caughley and Gunn 1996; Lande 1998). My previous studies assumed that such demographic disturbances were represented by proportional reductions of population size at a constant rate every generation (Tanaka 1997, 1998). The present study relies on an alternative assumption, that is, that the carrying capacity K decreases monotonically with time to a minimum value $K_{\min}$. An exponential function of time was employed for K , i.e., $K_t = (K_0 - K_{\min})(1 - k)^t + K_{\min}$, where k is the decreasing rate of K and K_0 is the initial carrying capacity. Thus, the decreasing rate of K slows down as K decreases and K approaches the asymptotic value $K_{\min}$. So long as the intrinsic rate of natural increase is larger than 0 ($r > 0$), the population size tracks K at any time, and also approaches $K_{\min}$ without going extinct. Extinction can occur only when the genetic load is inflated by inbreeding depression so that the intrinsic rate becomes negative ($r < 0$).

Environmental and genetic stochasticity

A real finite population in a stochastic environment is subject mainly to two kinds of stochasticity, environmental stochasticity and genetic stochasticity. The former results from temporal variation of any environmental factors and creates random fluctuation of the population growth rate. The latter, called random genetic drift, results from random samplings of gametes from a gene pool of parental generations and causes random dispersion of gene frequencies.

Environmental stochasticity was incorporated by an additional white noise term representing random fluctuations of the population growth rate. The carrying capacity, as well as the intrinsic rate of natural increase, is assumed to be

subject to environmental fluctuation (Feldman and Roughgarden 1975; Hanson and Tuckwell 1978; Iwasa 1998). Thus, the recurrence equation for population growth is $N_{t+1} = N_t \exp[r_{\max} (1 - N_t/K_t) + \varepsilon_t] \delta_{\text{inb}(t)}$, where ε_t is a random normal variate with mean 0 and variance σ_e^2 , $\delta_{\text{inb}(t)} = (1 - L_t)/(1 - \bar{L})$, and $L_t = ns\{\bar{q}_t^2 + F_t\bar{q}_t(1 - \bar{q}_t)\}$.

The random genetic drift in a multilocus system causes fluctuations of the mean gene frequency of all loci in addition to the dispersion of gene frequencies between loci. The dispersion of gene frequencies between loci was handled as increase in the variance of gene frequencies and the inbreeding coefficient. The effect of genetic drift on the mean gene frequency is incorporated by means of randomly sampling the mean gene frequency in the next generation from normal variates generated from the mean gene frequency after mutation and selection in the present generation. The mean gene frequency in the next generation, $\bar{q}_{t+1}$, is $\bar{q}_{t+1} = \bar{q}_t^* + \gamma_t$, where γ_t is a random normal variate with mean 0 and standard deviation $\sqrt{\bar{q}_t^*(1 - \bar{q}_t^*)/(2nN)}$, where $\bar{q}_t^*$ is the mean gene frequency after mutation and selection but before reproduction in the t th generation, N is the population size, and n is the number of loci.

Dynamics

The dynamics of the three parameters are summarized by the following joint recurrence equations:

$$\begin{bmatrix} \bar{q}_{t+1} \\ F_{t+1} \\ N_{t+1} \end{bmatrix} = \begin{bmatrix} \bar{q}_t - s\{\bar{q}_t^2 + F_t\bar{q}_t(1 - \bar{q}_t)\} + \mu + \gamma_t \\ \left\{ \frac{1}{2N_t} + \left(1 - \frac{1}{2N_t}\right)F_t \right\} (1 - 2\mu)(1 - s\bar{q}_t) \\ N_t \exp\left\{ r_{\max} \left(1 - \frac{N_t}{K_t}\right) + \varepsilon_t \right\} \delta_{\text{inb}(t)} \end{bmatrix} \quad (4)$$

Initial values of gene frequency, inbreeding coefficient, and population size are assumed to be equivalent to the stable equilibrium values without demographic disturbances. Iterative calculations were repeated until the population became extinct or persisted more than 1000 generations.

Numerical results

Because inbreeding depression is caused by decreased heterozygosity of recessive deleterious genes, the present analysis focused on recessive lethal genes rather than weakly deleterious genes with merely slight recessivity. From experiments using the balancer chromosomes of *Drosophila*, the genomic mutation rate of recessive lethals is approximately 0.03 (Crow and Simmons 1983; Woodruff et al. 1983; Eeken et al. 1987). I employed $\mu = 10^{-6}$ and $n = 15000$ throughout the analysis, which is compatible with the observed genomic mutation rate.

Extinction vortex by inbreeding depression

Numerical simulations using the deterministic model, in which we assumed $\sigma_e^2 = 0$ and $\text{var}(\gamma) = 0$, clarified relationships between the liability of extinction due to inbreeding depression and the population and genetic parameters except for the environmental variance of the population growth rate. We focused on how the liability of extinction vortex is influenced by different rates of demographic disturbance and the equilibrium population size.

With slow rates of population decline, the populations did not become extinct (Fig. 1; $k = 0.01$ and 0.02) within 1000 generations, although they did at fast rates (Fig. 1; $k = 0.04$ and 0.1), even if other parameter values were unaltered. These results are explained by the relative efficacy of purging selection against the deleterious genes compared to the rate of inbreeding depression. With slower rates of population decline, purging selection continued to act for a period long enough to exclude the deleterious genes. On the contrary, with higher rates of population decline, the positive feedback between inbreeding depression and declining population size exceeds the efficacy of purging selection, leading to an extinction vortex. The extremely high rates of decreasing population size in the final phase of extinction and the rapid increase of the inbreeding coefficient are compatible with the extinction vortex caused by interaction between these two elements (Fig. 1).

The equilibrium population size before the onset of demographic disturbances considerably influences the liability of the extinction vortex (Table 1). With larger equilibrium population sizes, larger numbers of deleterious genes are maintained per individual, and populations are more prone to extinction caused by inbreeding depression. Relatively small populations do not maintain a sufficient number of deleterious genes to induce the extinction vortex (Tanaka 1997).

The mutation-free intrinsic rate of population increase also influences the liability of the extinction vortex (Table 2). With the genetic parameter values employed in the analysis, the mutation-free net reproductive rate, $R_0 = e^{r_{\max}}$, equal to or larger than 1.3 never resulted in extinction within 1000 generations regardless of the disturbance rate.

Synergistic interaction between inbreeding depression and environmental stochasticity

With environmental and genetic stochasticity, simulations using an identical set of parameters can result in either extinction or persistence (Fig. 2). However, the probability of extinction is influenced by genetic and demographic parameters.

The environmental variance of the population growth rate inflates liability of extinction due to inbreeding depression (Fig. 3). Figure 3 exemplifies two cases with parameter values identical except for the environmental variance. The set of parameter values used in Fig. 3 does not result in extinction by an inbreeding vortex with the deterministic

Table 1. Results of deterministic simulations for various disturbance rates k and initial carrying capacities K_0 (equilibrium population size): parameter values are $\mu = 10^{-6}$, $n = 15000$, and $s = 1$

K_0	LE	Disturbance rate k															
		0.01	0.02	0.04	0.06	0.08	0.10	0.12	0.14	0.16	0.18	0.20	0.22	0.24	0.26	0.28	0.30
10^8	29.9	—	—	+	+	+	+	+	+	+	+	+	+	+	+	+	+
$10^{7.5}$	29.8	—	—	+	+	+	+	+	+	+	+	+	+	+	+	+	+
10^7	29.2	—	—	+	+	+	+	+	+	+	+	+	+	+	+	+	+
$10^{6.5}$	27.5	—	—	+	+	+	+	+	+	+	+	+	+	+	+	+	+
10^6	21.3	—	—	—	—	—	—	—	+	+	+	+	+	+	+	+	+
$10^{5.5}$	0.19	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
10^5	0.05	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
$10^{4.5}$	0.04	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
10^4	0.03	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

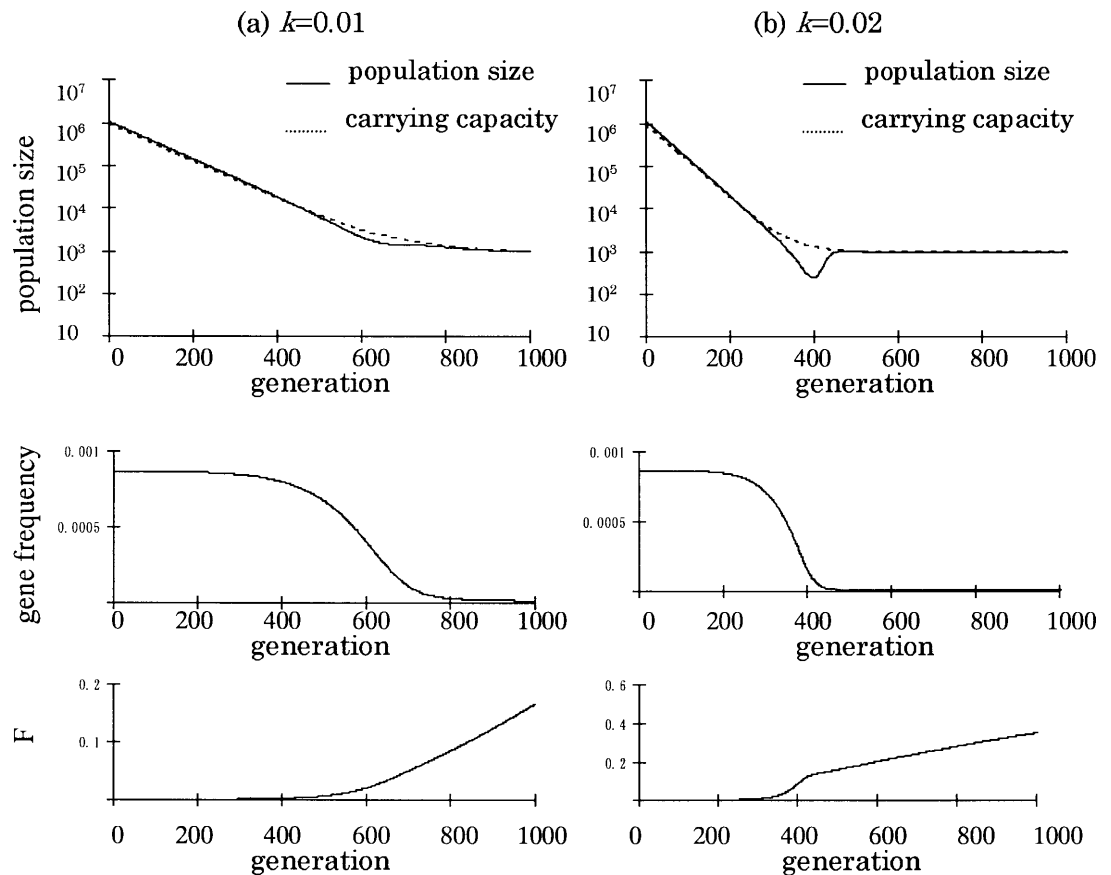
+, extinct within 1000 generations; —, persistent more than 1000 generations; LE, lethal equivalents at equilibrium

Table 2. Results of deterministic simulations for various disturbance rates k and net reproductive rates R_0 : parameter values are $\mu = 10^{-6}$, $n = 15000$, $K_0 = 10^6$, $K_{\min} = 10^3$, and $s = 1$

R_0	Disturbance rate															
	0.01	0.02	0.04	0.06	0.08	0.10	0.12	0.14	0.16	0.18	0.20	0.22	0.24	0.26	0.28	0.30
1.05	—	—	+	+	+	+	+	+	+	+	+	+	+	+	+	+
1.10	—	—	+	+	+	+	+	+	+	+	+	+	+	+	+	+
1.15	—	—	+	+	+	+	+	+	+	+	+	+	+	+	+	+
1.20	—	—	+	+	+	+	+	+	+	+	+	+	+	+	+	+
1.25	—	—	—	—	—	—	—	+	+	+	+	+	+	+	+	+
1.30	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
1.35	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
1.40	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

+, extinct within 1000 generations; —, persistent more than 1000 generations; R_0 , net reproductive rate at equilibrium

Fig. 1. Predictions with the deterministic model of population trajectories and changes in mean gene frequencies and inbreeding coefficients after the onset of demographic disturbances with four different disturbance rates, k (0.01, 0.02, 0.04, 0.1; **a–d**, respectively). Parameter values, except for the disturbance rate, are common among the simulations: $\mu = 10^{-6}$, $n = 15000$, $K_0 = 10^6$, $K_{\min} = 10^3$, $s = 1$



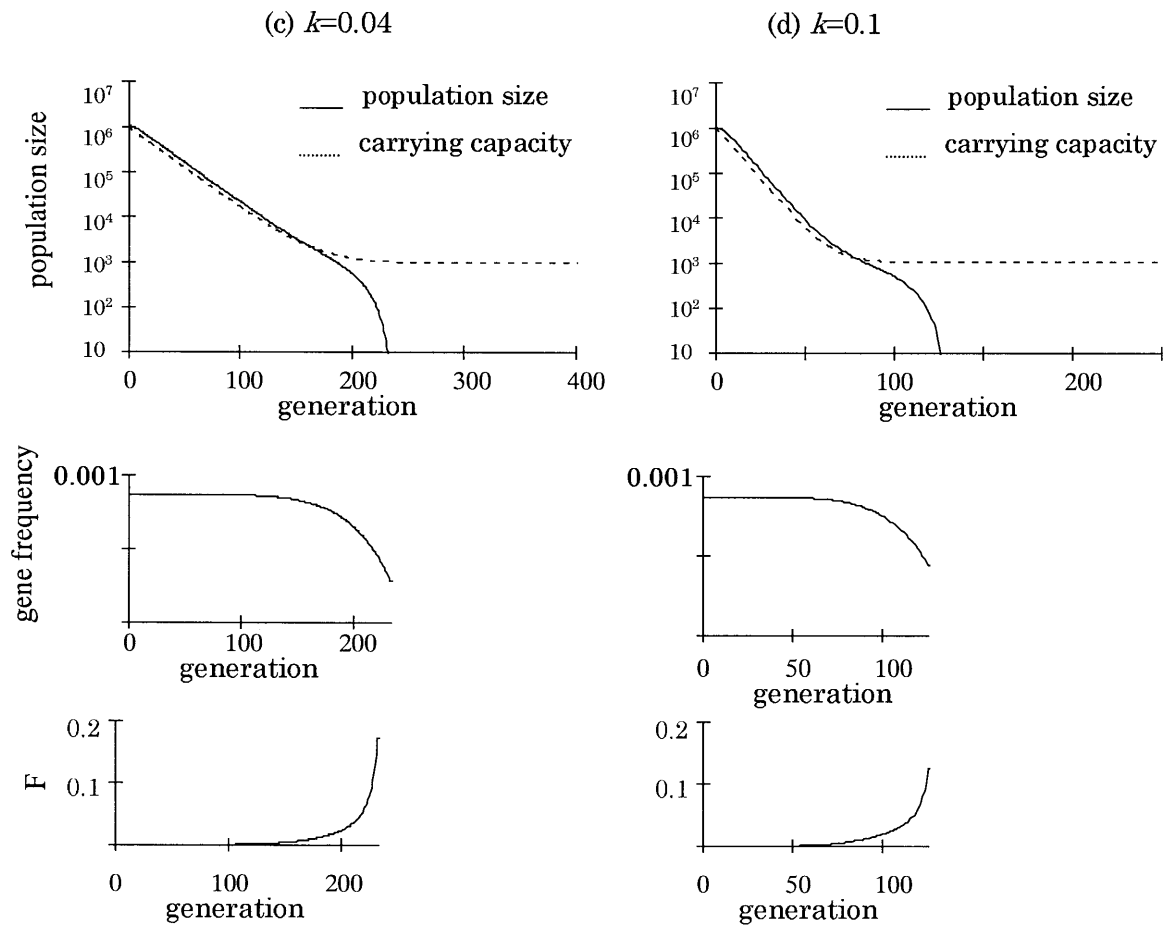


Fig. 1. Continued

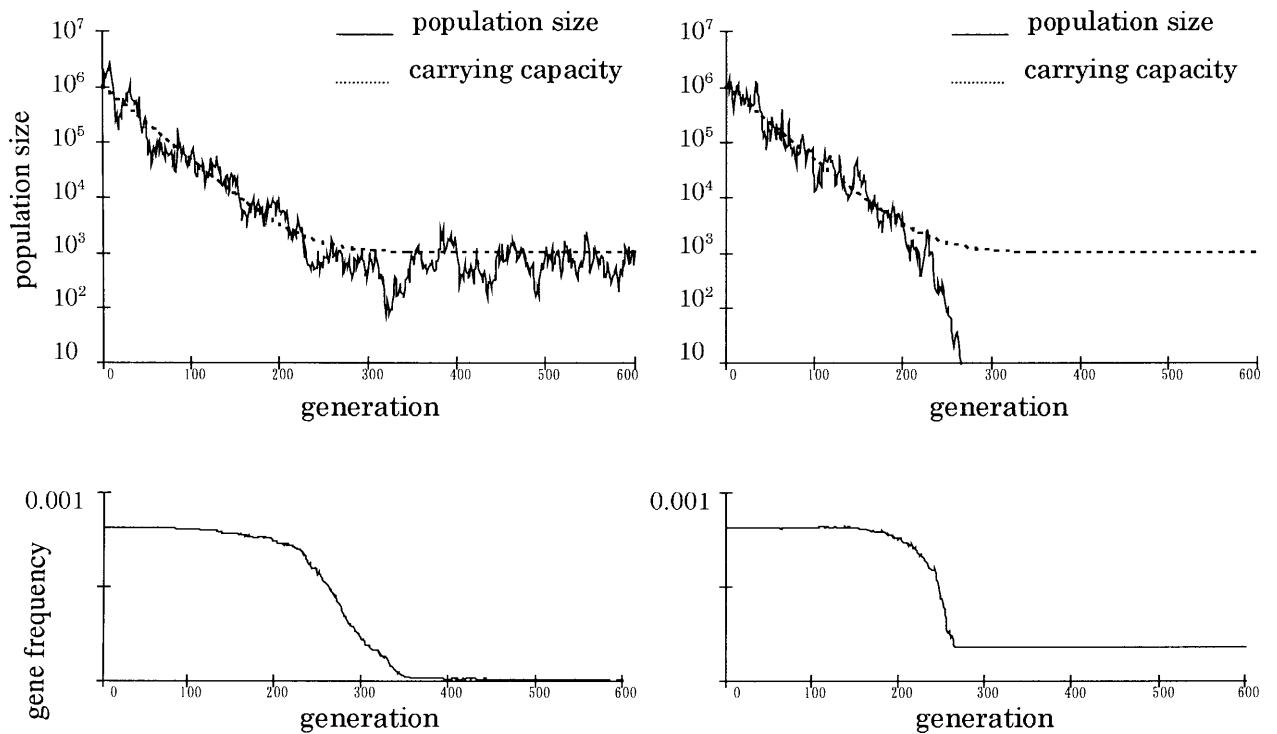


Fig. 2. Examples of population trajectories and gene frequency changes predicted with the stochastic model. All parameter values are equivalent between the two simulations: $\mu = 10^{-6}$, $n = 15000$, $K_0 = 10^6$, $K_{\min} = 10^5$, $k = 0.03$, $\nu = 0.1$, $s = 1$.

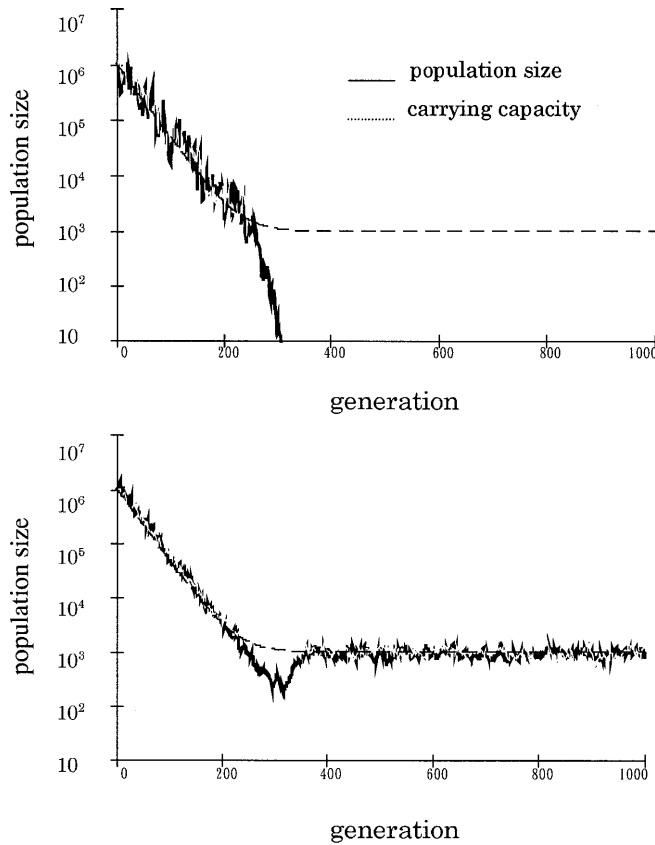


Fig. 3. Population trajectories exemplifying the influence of environmental stochasticity on liability of extinction by inbreeding depression. *Upper graph:* a large environmental variance of the intrinsic rate of natural increase ($v = 0.1$). *Lower graph:* a small variance ($v = 0.01$). All parameter values are equivalent between the two simulations: $\mu = 10^{-6}$, $n = 15000$, $K_0 = 10^6$, $K_{\min} = 10^3$, $k = 0.03$, $s = 1$

model ($\sigma_e^2 = 0$ and $\text{var}(\gamma) = 0$). To clarify the relationship between environmental stochasticity and liability of inbreeding vortex, for each set of parameters simulations were repeated for 300 runs of 400 generations, each of which was generated from different sets of random numbers for ε_t and γ_t (Fig. 4). The simulations were also repeated for different rates of demographic disturbance k . With environmental variances of population growth rate, even if they are small, an extinction risk due to the inbreeding vortex is generated for parameter sets that do not bring about extinction with the deterministic model. The larger the environmental variances, the more the extinction risk is inflated if the environmental variance is less than 0.05 (the coefficient of variation of r is 1.24). Thus, the results indicate that inbreeding depression interacts with environmental stochasticity to induce the extinction of monotonically decreasing populations.

The maximum environmental variance favoring extinction tended to slightly increase with the rate of demographic disturbance. The reason why the curves plotted for proportions of extinctions against environmental variances of r is inversely U-shaped is that the equilibrium effective population size considerably decreases with the very large environmental fluctuation of population size so that the populations

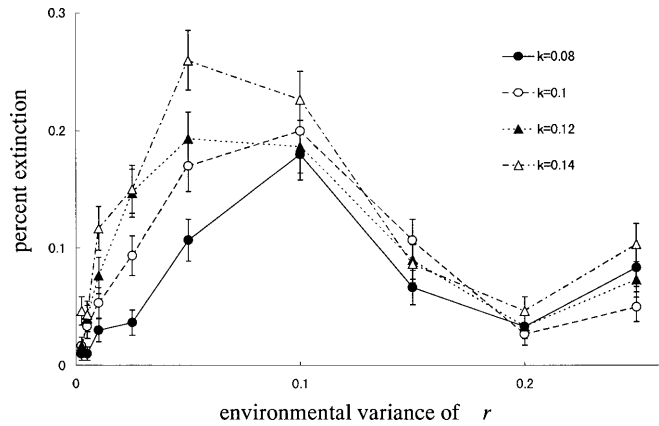


Fig. 4. Relationships between extinction probabilities caused by inbreeding depression and environmental variance of r . Each dot represents percent of extinction in 300 runs of the simulation for 400 generations (definition of extinction, $N < 10$). Different lines represent different rates of demographic disturbances; vertical bars, standard errors calculated from binomial distribution. Parameter values: $\mu = 10^{-6}$, $n = 15000$, $K_0 = 10^6$, $K_{\min} = 10^3$, $s = 1$

do not maintain a sufficient frequency of deleterious genes. The extinction rate rebounded when the environmental variance was larger than 0.2, presumably as the result of direct environmental stochasticity. Parallel simulations that excluded all genetics resulted in equally frequent extinctions when $\sigma_e^2 = 0.25$ (data not shown).

Discussion

The present analysis suggests that there is a synergistic interaction between inbreeding depression and environmental stochasticity that induces rapid extinction by inbreeding depression (inbreeding vortex). The simultaneous actions of the two factors can induce the inbreeding vortex even when a single action of either effect does not. The causal mechanism for this synergistic interaction is unclear. With environmental fluctuation of population size, an occasional reduction of population size may trigger the inbreeding vortex by escaping the effect of purging selection.

Liability to extinction caused by inbreeding depression under temporal demographic declines depends on the past history of populations. Because equilibrium gene frequencies of recessive deleterious genes (or numbers of lethal equivalents) are strongly dependent on the equilibrium population size, the population size at equilibrium before the onset of demographic disturbance greatly influences the liability of the inbreeding vortex (Tanaka 1997, 1998). In a small population, purging selection is so effective that very few deleterious genes can be maintained at equilibrium. If we discount other extinction factors important for small populations, e.g., accumulation of new deleterious mutations and demographic stochasticity (Caughley and Gunn 1996), a long-term small population may not be liable to short-term genetic extinction risk. It is reported, on the basis of offspring survivorship, that lethal equivalents are

very small in carnivores, including cheetahs, which presumably have smaller population numbers or experienced severe bottlenecks in the glacial period (Ralls et al. 1988; O'Brien et al. 1983, 1987).

The rate of population declines that are not caused by genetic factors is also important for inducing rapid extinction due to inbreeding depression. With slow rates of population decline, the efficacy of purging selection more than cancels out the effect of increased inbreeding. On the contrary, with a fast rate of decline, purging by selection cannot catch up with the effect of increased inbreeding.

Those results were derived from a standard population genetic model. However, they need empirical checks, preferably based on individual-based Monte Carlo simulations, in future work. Especially, the assumption of linkage equilibrium and the simplified description of stochastic dispersion of gene frequencies among loci may influence the results. Nonetheless, the main results in the present study provide some suggestions to practical conservation problems. Rapid extinction caused by inbreeding depression is likely when long-term large populations that have not experienced severe bottlenecks in the past rapidly decrease in population numbers due to anthropogenic factors. The environmental fluctuation of population size reinforces extinction from inbreeding depression. Evaluation of the minimum viable population for a monotonically declining large population should take into account the extinction risk aggravated by inbreeding depression.

Acknowledgments I thank Yasushi Harada, Yoh Iwasa, Mayuko Nakamaru, and Akira Sasaki for helpful discussion on the topic. This work is supported in part by CREST (Core Research for Evolutional Science and Technology) of Japan Science and Technology Corporation (JST) (principal investigator is J. Nakanishi).

References

- Awise JC, Hamrick JL (1996) Conservation genetics: case histories from nature. Chapman & Hall, New York
- Barrett SCH, Charlesworth D (1991) Effects of a change in the level of inbreeding on the genetic load. *Nature (Lond)* 352:522–524
- Britten HB (1996) Meta-analyses of the association between multilocus heterozygosity and fitness. *Evolution* 50:2158–2164
- Caughley G, Gunn A (1996) Conservation biology in theory and practice. Blackwell, Cambridge
- Crow JF, Kimura M (1970) An introduction to population genetics theory. Harper & Row, New York
- Crow JF, Simmons MJ (1983) The mutation load in *Drosophila*. In: Ashburner M, Carson HL, Thompson JN Jr (eds) The genetics and biology of *Drosophila*, vol. 3C. Academic Press, New York, pp 1–35
- David P, Delay B, Jarne P (1997) Heterozygosity and growth in the marine bivalve *Spisula ovalis*: testing alternative hypotheses. *Genet Res* 70:215–223
- Eeken JCJ, De Jong AWM, Green MM (1987) The spontaneous mutation rate in *Drosophila simulans*. *Mutat Res* 192:259–262
- Falconer DS (1989) Introduction to quantitative genetics, 3rd edn. Longman, New York
- Feldman MW, Roughgarden J (1975) A population's stationary distribution and chance of extinction in a stochastic environment with remarks on the theory of species packing. *Theor Popul Biol* 7: 197–207
- Frankel OH, Soule ME (1981) Conservation and evolution. Cambridge University Press, Cambridge
- Gilpin ME, Soule ME (1986) Minimum viable populations: the processes of species extinction. In: Soule ME (ed) Conservation biology: the science of scarcity and diversity. Sinauer, Sunderland, MA, pp 13–34
- Hanson FB, Tuckwell HC (1978) Persistence times of populations with large random fluctuations. *Theor Popul Biol* 14:46–61
- Iwasa Y (1998) Ecological risk assessment by the use of the probability of species extinction. In: Nakanishi J (ed) Proceedings of the 1st international workshop on risk evaluation and management of chemicals. Yokohama National University, Yokohama, pp 42–49
- Lande R (1998) Anthropogenic, ecological and genetic factors in extinction and conservation. *Res Popul Ecol* 40:259–269
- Loeschcke V, Tomiuk J, Jain SK (1994) Conservation genetics. Birkhauser, Basel
- Mitton JB (1993) Theory and data pertinent to the relationship between heterozygosity and fitness. In: Thornhill NW (ed) The natural history of inbreeding and outbreeding. University of Chicago Press, Chicago, pp 17–41
- Mitton JB, Grant MC (1984) Associations among protein heterozygosity, growth rate, and developmental homeostasis. *Annu Rev Ecol Syst* 15:479–499
- Nei M (1987) Molecular evolutionary genetics. Cambridge University Press, Cambridge
- O'Brien SJ, Wildt DE, Goldman D, Merrill DR, Bush M (1983) The cheetah is depauperate in genetic variation. *Science* 211:459–462
- O'Brien SJ, Wildt DE, Bush M, Caro TM, Fitzgibbon C, Aggundey I, Leakey RE (1987) East African cheetahs: evidence for two population bottlenecks? *Proc Natl Acad Sci USA* 84:508–511
- Ouborg NJ, Treuren RV (1994) The significance of genetic erosion in the process of extinction. IV. Inbreeding load and heterosis in relation to population size in the mint *Salvia pratensis*. *Evolution* 48: 996–1008
- Ralls K, Ballou J, Templeton A (1988) Estimates of lethal equivalents and the cost of inbreeding in mammals. *Conserv Biol* 2:185–193
- Saccheri I, Kuussaari M, Kankare M, Vikman P, Fortelius W, Hanski I (1998) Inbreeding and extinction in a butterfly metapopulation. *Nature (Lond)* 392:491–494
- Schierup MH (1998) The effect of enzyme heterozygosity on growth in a strictly outcrossing species, the self-incompatible *Arabidopsis petraea* (Brassicaceae). *Hereditas* 128:21–31
- Tanaka Y (1997) Extinction of populations due to inbreeding depression with demographic disturbances. *Res Popul Ecol* 39:57–66
- Tanaka Y (1998) Theoretical aspects of extinction by inbreeding depression. *Res Popul Ecol* 40:279–286
- van Noordwijk AJ (1994) The interaction of inbreeding depression and environmental stochasticity in the risk of extinction of small populations. In: Loeschcke V, Tomiuk J, Jain SK (eds) Conservation genetics. Birkhauser, Basel
- Willis JH (1993) Effects of different levels of inbreeding on fitness components in *Mimulus guttatus*. *Heredity* 69:562–572
- Woodruff RC, Slatko BE, Thompson JN Jr (1983) Factors affecting mutation rates in natural populations. In: Ashburner M, Carson HL, Thompson JN Jr (eds) The genetics and biology of *Drosophila*, vol. 3C. Academic Press, New York, pp 37–124
- Zouros E (1987) On the relationship between heterozygosity and heterosis: an evolution of evidence from marine mollusks. In: Rattari MC, Scandalios JG, Whitt GS (eds) Isozymes: current topics in biological and medical research, vol. 15. Liss, New York, pp 255–270